Exercise solutions- Lecture p.3

Exercise 1

Task A · Problem formulation

Scientific objective

Generate a Python analysis scaffold for calcium imaging data that loads recordings from multiple animals, computes ΔF/F summary measures for each stimulus condition, compares the two conditions, and produces one summary figure.

Unit of analysis

The primary unit of analysis is the animal. Within each animal, ΔF/F summaries are computed for the two stimulus conditions and then aggregated for cross-animal comparison.

Desired deliverable

A Python scaffold that:

- loads the data
- computes ΔF/F summaries by animal and condition
- compares the two conditions
- produces one figure
- is organized into clear functions with comments

What must not be assumed

- do not assume a specific file naming convention unless provided
- do not assume all animals have the same number of trials
- do not assume data are already cleaned
- do not assume a particular statistical test without being specified
- do not assume missing values are absent

Task B · Context block

File structure

Data are stored in one folder per animal. Each animal folder contains calcium imaging summary data for two stimulus conditions, stored as CSV files. Example:

```
data/
  animal_01/
```

```
    condition_A.csv
    condition_B.csv
  animal_02/
    condition_A.csv
    condition_B.csv
```

Variable names

Expected columns:

- `animal_id`
- `trial_id`
- `condition`
- `time`
- `fluorescence`
- `baseline_fluorescence`

Optional derived variable:

- `dff`

Available libraries

Use only:

- `pandas`
- `numpy`
- `matplotlib`

Expected outputs

- one function to load the data
- one function to compute ΔF/F
- one function to summarize ΔF/F by animal and condition
- one function to compare conditions
- one figure showing the condition-level comparison
- one output table with summary statistics

Task C · Validation block

- **One test:** Check that the computed ΔF/F column exists and contains numeric values for all non-missing rows.
- **One schema check:** Verify that the loaded data contain the required columns:
- **One failure condition:** Stop and report an error if one of the stimulus conditions is missing for any animal.
- **One required caveat:** The scaffold should not interpret condition differences as biologically meaningful unless the preprocessing, ΔF/F definition, and statistical assumptions have been verified by the researcher.

Actual prompt

```
You are helping generate a Python analysis scaffold for calcium
imaging data.

Objective:
Write Python code to load calcium imaging data from multiple animals
across two stimulus conditions, compute ΔF/F summary measures,
compare the two conditions, and produce one summary figure.

Unit of analysis:
The main unit of analysis is the animal. ΔF/F summaries should be
computed for each condition within each animal, then aggregated for
cross-animal comparison.

Context:
- Data are stored in one folder per animal.
- Each animal folder contains CSV files for two stimulus conditions.
- Example file structure:
  data/
    animal_01/
      condition_A.csv
      condition_B.csv
    animal_02/
      condition_A.csv
      condition_B.csv
- Expected columns are: animal_id, trial_id, condition, time,
fluorescence, baseline_fluorescence.
- A derived variable dff may be computed.
- Use only pandas, numpy, and matplotlib.

Constraints:
- Do not assume a specific file naming convention unless it is
explicitly provided.
- Do not assume all animals have the same number of trials.
- Do not assume the data are already cleaned.
- Do not assume a particular statistical test unless it is
explicitly specified.
- Do not assume missing values are absent.
- Do not fabricate files, columns, or outputs.
- If assumptions are required, state them explicitly.

Output:
1. A short plan
2. One function to load the data
3. One function to compute ΔF/F
4. One function to summarize ΔF/F by animal and condition
5. One function to compare the two conditions
```

```
6. One figure showing the condition-level comparison
7. One output table with summary statistics
8. Clear functions with brief comments

Validation requirements:
- Include one test that checks that the computed ΔF/F column exists
and contains numeric values for all non-missing rows.
- Include one schema check that verifies that the loaded data
contain the required columns.
- Include one failure condition that stops execution and reports an
error if one of the stimulus conditions is missing for any animal.
- Include one caveat stating that condition differences should not
be interpreted as biologically meaningful unless the preprocessing,
ΔF/F definition, and statistical assumptions have been verified by
the researcher.
```

Exercise 2

Task A · Problem formulation

Scientific objective

Generate a Python analysis scaffold for spike train data that loads recordings from multiple neurons, computes firing-rate summaries for rest and movement conditions, compares the two conditions, and produces one summary figure.

Unit of analysis

The primary unit of analysis is the **neuron**. Within each neuron, firing-rate summaries are computed separately for rest and movement, then compared across neurons.

Desired deliverable

A Python scaffold that:

- loads spike train data
- computes firing rates by neuron and condition
- compares rest versus movement
- produces one figure
- is organized into clear functions with brief comments

What must not be assumed

- do not assume all neurons have the same recording duration
- do not assume every neuron has data in both conditions unless specified
- do not assume spike times are already cleaned
- do not assume a specific statistical test without being instructed
- do not assume file names, bin sizes, or trial structure unless provided

---

Task B · Context block

File structure

Data are stored in one CSV file per recording session.

```
data/
      session_01.csv
      session_02.csv
      session_03.csv
```

Variable names

Expected columns:

- `session_id`
- `neuron_id`
- `condition`

- `spike_time_ms`
- `trial_id`

Optional metadata columns:

- `animal_id`
- `recording_duration_ms`

Available libraries
Use only:

- `pandas`
- `numpy`
- `matplotlib`

Expected outputs

- one function to load the data
- one function to compute firing rates
- one function to summarize firing rates by neuron and condition
- one function to compare the two conditions
- one figure showing the condition-level comparison
- one summary table with per-neuron results

Task C · Validation block

- **One test**: Check that the firing-rate output exists and is numeric.
- **One schema check**: Verify that the loaded data contain the required columns
- **One failure condition**: Stop and report an error if one of the two conditions is missing for all neurons in a session.
- **One required caveat**: The scaffold should not interpret a difference in firing rate as evidence of a specific neural mechanism unless the experimental design, preprocessing choices, and statistical assumptions have been verified by the researcher.

Actual prompt

```
You are helping generate a Python analysis scaffold for spike train
data.

Objective:
Write Python code to load spike train data from multiple neurons
recorded during two behavioral conditions, rest and movement,
compute firing-rate summaries, compare the two conditions, and
produce one summary figure.

Unit of analysis:
The main unit of analysis is the neuron. Firing-rate summaries
should be computed separately for rest and movement within each
neuron, then compared across neurons.

Context:
- Data are stored in one CSV file per recording session.
- Example file structure:
  data/
    session_01.csv
    session_02.csv
    session_03.csv
- Expected columns are: session_id, neuron_id, condition,
spike_time_ms, trial_id.
- Optional metadata columns may include: animal_id,
recording_duration_ms.
- Use only pandas, numpy, and matplotlib.

Constraints:
- Do not assume all neurons have the same recording duration.
- Do not assume every neuron has data in both conditions unless this
is confirmed by the data.
- Do not assume spike times are already cleaned.
- Do not assume a specific statistical test unless explicitly
instructed.
- Do not assume file names, bin sizes, or trial structure unless
provided.
- Do not fabricate files, columns, or results.
- If assumptions are required, state them explicitly.

Output:
1. A short plan
2. One function to load the data
3. One function to compute firing rates
4. One function to summarize firing rates by neuron and condition
5. One function to compare rest versus movement
6. One figure showing the condition-level comparison
```

```
7. One summary table with per-neuron results
8. Brief comments in the code

Validation requirements:
- Include one test that checks that the firing-rate output exists
and is numeric.
- Include one schema check that verifies that the loaded data
contain the required columns.
- Include one failure condition that stops execution and reports an
error if one of the two conditions is missing for all neurons in a
session.
- Include one caveat stating that any difference in firing rate
should not be interpreted as evidence of a specific neural mechanism
unless the experimental design, preprocessing choices, and
statistical assumptions have been verified by the researcher.
```